	Target engagement	Dose	Efficacy	Safety
Hem.	**PNH** anti-C5 antibodies (4)	**2022** Hemolysis assay with C5 R885H variant	**2020** Cynomolgus PK/PD (crovalimab)	**2018** Complement hemolysis assay in human serum	**2020** Drug–target–drug complex assay, human C5
Met.	**hyperlipidemia** CETP inhibitors (3)	**2014** Human CETP/apoA-I transgenic mouse	**2014** CETP/apoA-I mouse; rat blood pressure ■ Exposure far below prediction	**2011** Human CETP loss-of-function genotypes □ Outcome trial negative	**2014** Rat blood pressure + human adrenal H295R cells
Psy.	**insomnia** HCRTR1/2 antagonists (6)	**2013** Rat, dog and rhesus sleep EEG/EMG (suvorexant)	**2022** PBPK-PD model of brain OX2 receptor occupancy (daridorexant, 65% occupancy threshold)	**2012** Rat, dog and rhesus sleep EEG/EMG (suvorexant)	**2019** Mouse emotion-induced cataplexy (lemborexant) ■ Animals no abuse; humans liked it
Respiratory	**cystic fibrosis** CFTR correctors (10)	**2024** FRT cells expressing CFTR variants ■ Third corrector added nothing	**2015** Rat and dog toxicology (lumacaftor)	**2024** FRT cells expressing CFTR variants	**2025** Rat and dog tox (vanzacaftor)
	cystic fibrosis CFTR potentiators (5)	**2026** Patient-derived primary nasal epithelial cells (HNE, Ussing chamber theratyping)	**2024** Patient-derived HBE cells (Ussing chamber)	**2024** FRT cells carrying each CFTR variant (Ussing chamber)	**2024** Rat toxicology (juvenile and adult, ivacaftor program)
Uro.	**interstitial cystitis** anti-NGF antibodies (3)	**1997** Rodent chemical cystitis	**1997** Rodent chemical cystitis	**2022** ASP6294 vs anti-NGF IgG in arthritic rodents ASP6294 missed IC pain endpoint	**2012** Primate and rodent joint tox Joint harm missed; IC trials put on hold
Other	**atopic eczema** PDE4 inhibitors (5)	**2019** Stimulated human blood cells (PBMC cytokine release)	**2016** Minipig dermal toxicity (3- and 9-month)	**2023** Mouse chronic allergic contact dermatitis (difamilast) ■ Explant engaged; primary failed	**2016** Rabbit primary skin irritation